Jaymin Sheladia

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Education

University of Southern California — Los Angeles, USA
Master of Science in Computer Science — January 2026 – December 2027

Graphic Era Deemed to be University — Dehradun, India
B.Tech in Computer Science and Engineering — October 2021 – June 2025

Technical Skills

Languages: C, C++, Python, Java, JavaScript, HTML, SQL
Frameworks/Libraries: Angular, React, Express.js, Spring Boot, Flask, TensorFlow, Keras, OpenCV, Scikit-learn
Developer Tools: Git, GitHub Actions, Docker, VS Code, PyCharm, Jupyter Notebook, Postman
Databases: MySQL, MongoDB, PostgreSQL
Concepts: Data Structures & Algorithms, Distributed Systems, Unix/Linux, RESTful API Design, CI/CD
Relevant Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Applied Natural Language Processing, Agentic AI Systems

Work Experience

Cognizant Technology Solutions — May 2025 – September 2025
Programmer Analyst Trainee – Software Engineering — Coimbatore, India

- Engineered a full-stack Travel Planner and Booking System, designing **10+** RESTful APIs in Spring Boot and Angular to modernize manual booking workflows and centralize itinerary, booking, and preference management.
- Integrated a Generative AI-based recommendation engine that analyzes **3** key factors – user interests, budget, and trip duration – to surface personalized suggestions that simplified the booking decision process.

CODTECH IT SOLUTIONS — June 2024 – August 2024
Machine Learning Intern — Remote

- Designed and trained a fraud detection model to flag fraudulent transactions, applying anomaly detection and supervised learning on over **200,000** credit card transactions to reach **98%** accuracy.
- Optimized algorithms and rebuilt data preprocessing pipelines for existing production models, enhancing their accuracy by **12–15%** across multiple live machine learning projects and reporting dashboards.

Projects

AI Research Assistant | GitHub | FastAPI, Next.js, Gemini API — June 2026 – July 2026

- Architected a full-stack research assistant with a FastAPI/PostgreSQL (pgvector) backend and Next.js frontend, building RAG-based paper chat, citation-aware related-work search, and a knowledge graph across ingested papers.
- Benchmarked the ingestion pipeline against live Gemini and Voyage API calls, achieving end-to-end processing in under **4.3** seconds per paper, with LLM calls accounting for **85–95%** of total latency.

TechHire: AI-Powered Job Search Platform | GitHub | React, FastAPI, PostgreSQL — May 2026 – June 2026

- Built a full-stack job search platform with a React/Vite frontend and FastAPI/PostgreSQL backend to simplify job discovery, implementing filters across **6** dimensions including skills, salary, and visa sponsorship.
- Leveraged the Groq API to auto-generate AI-powered job summaries and built resume upload, text-extraction, and resume-to-job comparison features backed by automated Python scrapers pulling live listings.

Sequence Alignment: DP vs. Divide & Conquer | GitHub | Python — March 2026 – April 2026

- Implemented and benchmarked the classical Needleman-Wunsch dynamic programming approach against Hirschberg's memory-efficient divide-and-conquer algorithm for large genomic sequence inputs.
- Reduced memory usage from $\mathbf{O(m \times n)}$ to $\mathbf{O(m+n)}$ using Hirschberg's algorithm, cutting peak memory consumption by up to **90%** on large inputs while preserving identical optimal alignments.

Research

Deep Learning Approach for Malicious URL Detection | Co-author; Published Research Paper — 2025

- Compared **4** deep learning architectures (CNN, RNN, LSTM, Bi-LSTM) for malicious URL detection, engineering character-level preprocessing pipelines on the IEEE Dataport dataset for sequential analysis at scale.
- Achieved **99.25%** accuracy with the Bi-LSTM model, outperforming CNN (**86.5%**), LSTM (**63.75%**), and RNN (**53%**) baselines across standard classification and error-rate metrics.